

ECSE 318 Lab 1 Group 9

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All files, responses, and testbench outputs can be found in their respective folders.

For convenience free response questions are also posted here.

Question 1 :

Part C

No. ModelSim can only prove that for any given input the two models are equivalent.

To prove they are exactly the same you would have to test every single set of inputs which gets unfeasible at scale.

Question 2:

Part 2:

The CLA consists of an AND gate, OR gate, and XOR gate. Each gate has $t_{10} = 10\text{ns}$ of delay so we can calculate the worst case delay is 30ns .

Part 3:

As seen from Q2Output.png, the worst case propagation happens when we are adding c_{in} to flip the MSB of sum (in this case $S[3]$). We see that this value is $2030000\text{ps} - 2000000\text{ps} = 3000\text{ps} = 30\text{ns}$.

Question 3

Adding a sequence of 10 8-bit binary numbers using a CSA design uses an 8 stage CSA followed by a regular adder if all the 3 inputs adders computed their outputs sequentially.

However, if all the 3 input adders computes their CSA outputs in parallel, then a 5 stage CSA followed by a regular adder can be used to add the sequence of numbers.

This is because the initial 10 numbers can be group into sets of 3s with the remainder number temporarily unattended, then each sets of 3 can have their inputs summed in a CSA stage resulting in 7 remaining operands (3 sums, 3 carry outs and the remaining 10th number).

If this process is recursively performed for the 7 operands and their CSA stage outputs then then number of operands after all the parallel CSA stage changes from 10 to 7 to 5 to 4 to 3 to 2 showing the 5 stages.

Question 4

Since the circuit is combinational, the time it takes to complete one operation is determined by the longest path through the circuit,

which we can trace through the number of gates it needs to go through.

Per the problem, we assume each gate has a one unit of delay

We can look at this cell by cell by looking at how many gates there are in one CAS:

In fa.v, we have the following:

```
assign cout = (a & b) | (a & cin) | (b & cin);  
assign sum = a ^ b ^ cin;
```

Both cout and sum go through two gates. Therefore, we have two units of delay per fa. We have one fa in each cas cell and nothing else. So each cas cell goes through 4 units of delay.

We have four of these 4 cells per row, so that ends of being $2 * 4$ units of delay from the cas cells in one row. Additionally, in each cell, we have a xor gate (assign b_control = m ^ b;) which runs in parallel per row. Therefore, we add a +1 unit delay per row, to give us $8 + 1 = 9$ units of delay per row. Since we have 5 of these rows, we have $9 * 5 = 45$ units of delay.

Question 5

Since the operations are sequential, we consider both the setup/hold times of the flip flops as well as the propagation delay of the logic gates. Specifically, we consider the longest logical path to find what the worst case delay is.

There are two logical components that contribute to propagation delay in our implementation of the divider

```
assign R_N = (count == 1) ? R : {R[N-1:0], D[N-1]}; // Mux
```

The delay for a 2-to-1 MUX in the datapath is 2 gate delays.

assign sum = control ? (R_N - Mwide) : (R_N + Mwide); // fa

The delay of a 5 bit full adder is the delay of a normal full adder x 5 maximum carries. since each full adder is 2 gates (as shown in Q4), we can say that the propagation delay is $2 * 5 = 10$.

Add onto the setup time of 0.5 units, we have a total time to complete one cycle of $2 + 10 + 0.5 = 12.5$.

Since we need 4 cycles to complete a division, we get a total time to complete one operation of $12.5 * 5 = 62.5$.